

Research Statement

Distributed Abductive Reasoning for Self-Explaining Digital Twins

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1 Introduction and Motivation

As digital twins transition from centralized simulations to distributed cyber-physical companions, they must handle the "reality gap" between model predictions and real-world observations. Current systems often treat these discrepancies as numerical errors to be minimized, without providing a human-interpretable explanation of their root causes. This research statement proposes a hierarchical and distributed explanation architecture based on abductive reasoning, enabling digital twins to interpret discrepancies as actionable knowledge about the system state and its requirements.

2 Current Research: Physics-Informed Digital Twins

In my current work at Sonatrach, I have developed a generative surrogate for gas turbine monitoring using **Mamba-SSM** and **xLSTM** architectures. A key achievement of this work is the deployment of these complex hybrid models onto **Edge devices**, where real-time performance and local autonomy are critical. However, my experience has shown that local model adaptation is insufficient if the digital twin cannot "explain" why a mismatch occurred—whether due to sensor drift, environmental shifts, or physical degradation. My background in edge-deployed AI provides a solid foundation for building the distributed reasoning layers needed for explainable twins.

3 Proposed Research Directions

Building upon the objectives of the **Distributed AI** project at Université Côte d’Azur, I propose to investigate the following axes:

3.1 1. Abductive Reasoning for Hybrid Models

Unlike deduction or induction, abductive reasoning focuses on generating the most plausible explanations for unexpected observations. I propose to develop abductive mechanisms that can bridge the gap between physics-based equations and data-driven learning. By formalizing the "surprise" encountered by a digital twin, we can generate hypotheses that distinguish between internal model deficiencies and external physical disturbances.

3.2 2. Hierarchical Edge-Fog-Cloud Explanation Architecture

I intend to investigate a distributed reasoning framework where low-level anomalies are detected at the **Edge**, refined into local explanations at the **Fog** level, and finally validated against global system requirements in the **Cloud**. This multi-level approach ensures that explanations are contextual and traceable, providing stakeholders with different granularities of insight depending on their roles.

3.3 3. Requirement-Aware Self-Explanation

A critical challenge in self-explaining systems is ensuring that explanations remain consistent with the broader system requirements. I propose to integrate "requirement-aware" constraints into the abductive process, ensuring that the hypotheses generated by the digital twin do not violate safety or performance boundaries.

4 Alignment with the EDT Program

My background in distributed AI and industrial implementation aligns with the goal of creating a national platform for digital twin engineering. I am eager to contribute to the ****Artemis platform**** by providing a distributed reasoning layer that enhances the transparency and trustworthiness of hybrid digital twins across the consortium.

5 Conclusion

The goal of my PhD will be to move beyond reactive error correction towards proactive, self-explaining digital twins. By combining abductive reasoning with distributed architectures, I aim to provide the foundational tools necessary for the next generation of transparent and trustworthy cyber-physical companions.